H10: Potentials and fields

10.1 the potential formulation

scalar V and vector A potentials

we can still define: $\mathbf{B} = \nabla \times \mathbf{A}$,
however, last time for V we relied on ∇xE, which is disproven due to:

$$\nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t},$$

thus in a changing magnetic field E != -∇V
However:

$$\nabla \times \mathbf{E} = -\frac{\partial}{\partial t}(\nabla \times \mathbf{A}),$$

<=>

$$\nabla \times \left(\mathbf{E} + \frac{\partial \mathbf{A}}{\partial t}\right) = \mathbf{0}.$$

thus, we can define:

$$\mathbf{E} = -\nabla V - \frac{\partial \mathbf{A}}{\partial t}.$$

new Poisson's equation

in a non-static case we thus have to account for the new potential formulation:

$$\nabla^2 V + \frac{\partial}{\partial t}(\nabla \cdot \mathbf{A}) = -\frac{1}{\epsilon_0}\rho;$$

which reduces to Poisson's equation in static case

relation A,V and J
eq10.5

we can put the relations for B and E in Maxwell's equation to find:

$$\nabla \times (\nabla \times \mathbf{A}) = \mu_0 \mathbf{J} - \mu_0 \epsilon_0 \nabla \left(\frac{\partial V}{\partial t}\right) - \mu_0 \epsilon_0 \frac{\partial^2 \mathbf{A}}{\partial t^2},$$

using vector identity ∇x(∇xA)=∇(∇.A)-∇²A and rearrange a bit to find:

$$\left(\nabla^2 \mathbf{A} - \mu_0 \epsilon_0 \frac{\partial^2 \mathbf{A}}{\partial t^2}\right) - \nabla \left(\nabla \cdot \mathbf{A} + \mu_0 \epsilon_0 \frac{\partial V}{\partial t}\right) = -\mu_0 \mathbf{J}.$$

10.1.2 Gauge transformations

Gauge transformations

suppose we have two potentials (V,**A**) and (V',**A'**) for the *same* elec. and magn. fields
> now they differ by a factor α and β:

$$\mathbf{A}' = \mathbf{A} + \boldsymbol{\alpha} \quad \text{and} \quad V' = V + \beta.$$

since the **A** and **A'** produce the same magnetic field, their curls have to be the same:

$$\nabla \times \boldsymbol{\alpha} = \mathbf{0}.$$

therefore **α** is the gradient of some scalar λ:

$$\boldsymbol{\alpha} = \nabla \lambda.$$

now, V and V' also produce the same electric field, thus:

$$\nabla \beta + \frac{\partial \boldsymbol{\alpha}}{\partial t} = \mathbf{0},$$

or thus:

$$\nabla \left(\beta + \frac{\partial \lambda}{\partial t}\right) = \mathbf{0}.$$

the term inside the parenthesis is therefore time-independent, and can be written as:

$$\beta = -\frac{\partial \lambda}{\partial t} + k(t).$$

where we can absorb k(t) inside λ, which wont affect the gradient of λ
> we therefore find the Gauge transformations:

$$\left.\begin{aligned} \mathbf{A}' &= \mathbf{A} + \nabla \lambda, \\ V' &= V - \frac{\partial \lambda}{\partial t}. \end{aligned}\right\}$$

10.1.3 Coulomb gauge and Lorentz gauge

Coulomb gauge

The defining choice for this gauge is:

$$\nabla \cdot \mathbf{A} = 0.$$

from this, we see that Poisson's equation becomes:

$$\nabla^2 V = -\frac{1}{\epsilon_0}\rho.$$

we can see that if we change V, the field changes immediately
ie: if i change V in the lab, V on the moon will change instantly
> doesn't seem to hold with relativity
> fear not: V itself isn't a measurable quantity, we can only measure E, which also depends on A
> -∇V-∂A/∂t doesn't change instantly

Lorenz gauge

the defining choice for this gauge is:

$$\nabla \cdot \mathbf{A} = -\mu_0\epsilon_0\frac{\partial V}{\partial t}$$

this eliminates the middle term in eq10.5:

$$\nabla \cdot \mathbf{A} = -\mu_0\epsilon_0\frac{\partial V}{\partial t}.$$

meanwhile, Poisson's equation becomes:

$$\nabla^2 V - \mu_0\epsilon_0\frac{\partial^2 V}{\partial t^2} = -\frac{1}{\epsilon_0}\rho.$$

we can define the transformation, d'Alembertian:

$$\nabla^2 - \mu_0\epsilon_0\frac{\partial^2}{\partial t^2} \equiv \Box^2, \quad \Rightarrow \quad \text{(i)} \quad \Box^2 V = -\frac{1}{\epsilon_0}\rho,$$

$$\text{(ii)} \quad \Box^2 \mathbf{A} = -\mu_0 \mathbf{J}.$$

>> this forms an inhomogeneous wave equation

10.1.4 Lorentz force law in potential form

Lorents force law in potential

express Lorentz force in terms of potentials:

$$\mathbf{F} = \frac{d\mathbf{p}}{dt} = q(\mathbf{E} + \mathbf{v} \times \mathbf{B}) = q\left[-\nabla V - \frac{\partial \mathbf{A}}{\partial t} + \mathbf{v} \times (\nabla \times \mathbf{A})\right]$$

with product rule $\nabla(\mathbf{v} \cdot \mathbf{A}) = \mathbf{v} \times (\nabla \times \mathbf{A}) + (\mathbf{v} \cdot \nabla)\mathbf{A}$:

$$\frac{d\mathbf{p}}{dt} = -q\left[\frac{\partial \mathbf{A}}{\partial t} + (\mathbf{v} \cdot \nabla)\mathbf{A} + \nabla(V - \mathbf{v} \cdot \mathbf{A})\right]$$

now define the convective derivative of **A** as:

$$\left[\frac{\partial \mathbf{A}}{\partial t} + (\mathbf{v} \cdot \nabla)\mathbf{A}\right]$$

> represents the rate of change of A at the moving location of the particle

convective derivative of A

suppose at time t the particle is at point **r** with potential **A**(**r**,t)
> a moment dt later it's at **r**+**v**dt with potential **A**(**r**+**v**dt, t+dt)
> the change in **A** is then:

$$d\mathbf{A} = \mathbf{A}(\mathbf{r} + \mathbf{v}\,dt, t + dt) - \mathbf{A}(\mathbf{r}, t)$$
$$= \left(\frac{\partial \mathbf{A}}{\partial x}\right)(v_x\,dt) + \left(\frac{\partial \mathbf{A}}{\partial y}\right)(v_y\,dt) + \left(\frac{\partial \mathbf{A}}{\partial z}\right)(v_z\,dt) + \left(\frac{\partial \mathbf{A}}{\partial t}\right)dt,$$

thus:

$$\frac{d\mathbf{A}}{dt} = \frac{\partial \mathbf{A}}{\partial t} + (\mathbf{v} \cdot \nabla)\mathbf{A}.$$

>> as the particle move, the potential changes because of: - variation in time
- variation in space

canonical momentum p_{can}

the force law reads: $\frac{d}{dt}(\mathbf{p} + q\mathbf{A}) = -\nabla\left[q(V - \mathbf{v} \cdot \mathbf{A})\right].$

which looks a lot like the formula in mechanics: d**p**/dt = -∇U
> therefore define: $\mathbf{p}_{\text{can}} = \mathbf{p} + q\mathbf{A},$

and: $U_{\text{vel}} = q(V - \mathbf{v} \cdot \mathbf{A}).$

10.2 continuous distributions	
10.2.1 retarded potentials	
retarded time	For a moving charge, its not the status of the source right now that matters > rather, it's the condition at an earlier time t_r : $$t_r \equiv t - \frac{\imath}{c}.$$
retarded potentials	We have to account for the retarded time in nonstatic sources: $$V(\mathbf{r},t) = \frac{1}{4\pi\epsilon_0}\int \frac{\rho(\mathbf{r}',t_r)}{\imath}\,d\tau', \qquad \mathbf{A}(\mathbf{r},t) = \frac{\mu_0}{4\pi}\int \frac{\mathbf{J}(\mathbf{r}',t_r)}{\imath}\,d\tau'.$$ >> proof: see p451
advanced time and potentials	define: $$t_a \equiv t + \frac{\imath}{c}.$$ and the potentials: $$V_a(\mathbf{r},t) = \frac{1}{4\pi\epsilon_0}\int \frac{\rho(\mathbf{r}',t_a)}{\imath}\,d\tau', \qquad \mathbf{A}_a(\mathbf{r},t) = \frac{\mu_0}{4\pi}\int \frac{\mathbf{J}(\mathbf{r}',t_a)}{\imath}\,d\tau',$$ >> this corresponds completely with the proof given on p451 > however: its in violation with the principle of causality nl: it implies that the effect precedes the cause > this does show however that there is no distinguish made between past and future
10.2.2 Jefimenko's equations	
time retarded E	given the retarded potentials: $$V(\mathbf{r},t) = \frac{1}{4\pi\epsilon_0}\int \frac{\rho(\mathbf{r}',t_r)}{\imath}\,d\tau', \qquad \mathbf{A}(\mathbf{r},t) = \frac{\mu_0}{4\pi}\int \frac{\mathbf{J}(\mathbf{r}',t_r)}{\imath}\,d\tau',$$ we know $\mathbf{E} = -\nabla V - \frac{\partial \mathbf{A}}{\partial t}$, with $\frac{\partial \mathbf{A}}{\partial t} = \frac{\mu_0}{4\pi}\int \frac{\dot{\mathbf{J}}}{\imath}\,d\tau'.$ we then have: $\mathbf{E}(\mathbf{r},t) = \frac{1}{4\pi\epsilon_0}\int \left[\frac{\rho(\mathbf{r}',t_r)}{\imath^2}\hat{\boldsymbol{\imath}} + \frac{\dot{\rho}(\mathbf{r}',t_r)}{c\imath}\hat{\boldsymbol{\imath}} - \frac{\dot{\mathbf{J}}(\mathbf{r}',t_r)}{c^2\imath}\right]d\tau'.$
time retarded B	the curl of A: $\nabla\times\mathbf{A} = \frac{\mu_0}{4\pi}\int\left[\frac{1}{\imath}(\nabla\times\mathbf{J}) - \mathbf{J}\times\nabla\left(\frac{1}{\imath}\right)\right]d\tau'.$ with $(\nabla\times\mathbf{J})_x = \frac{\partial J_z}{\partial y} - \frac{\partial J_y}{\partial z},$ and $\frac{\partial J_z}{\partial y} = \dot{J}_z\frac{\partial t_r}{\partial y} = -\frac{1}{c}\dot{J}_z\frac{\partial \imath}{\partial y},$ so: $(\nabla\times\mathbf{J})_x = -\frac{1}{c}\left(\dot{J}_z\frac{\partial \imath}{\partial y} - \dot{J}_y\frac{\partial \imath}{\partial z}\right) = \frac{1}{c}\left[\dot{\mathbf{J}}\times(\nabla\imath)\right]_x.$ but $\nabla\imath = \hat{\boldsymbol{\imath}}$, so: $\nabla\times\mathbf{J} = \frac{1}{c}\dot{\mathbf{J}}\times\hat{\boldsymbol{\imath}}.$ hence: $\mathbf{B}(\mathbf{r},t) = \frac{\mu_0}{4\pi}\int\left[\frac{\mathbf{J}(\mathbf{r}',t_r)}{\imath^2} + \frac{\dot{\mathbf{J}}(\mathbf{r}',t_r)}{c\imath}\right]\times\hat{\boldsymbol{\imath}}\,d\tau'.$

10.3 point charges	
10.3.1 Liénard-Wiechert potentials	
potential V of a moving point charge	suppose we have a moving point charge with velocity **w**(t) > we can find V with: $$V(\mathbf{r},t)=\frac{1}{4\pi\epsilon_0}\int\frac{\rho(\mathbf{r}',t_r)}{\imath}\,d\tau'$$ however, we need to account for the retarded time nl: to calculate the total charge of a configuration, we must integrate ρ over the entire distribution at one instant of time > here, t_r=t-r/c obliges us to evaluate ρ at different times for diff. parts of the configuration > this gives us an extra factor $(1-rv/c)^{-1}$: $$\int\rho(\mathbf{r}',t_r)\,d\tau'=\frac{q}{1-\hat{\boldsymbol{\imath}}\cdot\mathbf{v}/c}.$$ which gives us: $$V(\mathbf{r},t)=\frac{1}{4\pi\epsilon_0}\frac{qc}{(\imath c-\boldsymbol{\imath}\cdot\mathbf{v})},$$ with **v** the velocity at the retarded time and $\boldsymbol{\imath}$ the vector from the retarded position to the field point **r** >> proof see p457!!
A for a moving point charge	we find for A: $\mathbf{A}(\mathbf{r},t)=\frac{\mu_0}{4\pi}\int\frac{\rho(\mathbf{r}',t_r)\mathbf{v}(t_r)}{\imath}\,d\tau'=\frac{\mu_0}{4\pi}\frac{\mathbf{v}}{\imath}\int\rho(\mathbf{r}',t_r)\,d\tau',$ or thus: $\mathbf{A}(\mathbf{r},t)=\frac{\mu_0}{4\pi}\frac{qc\mathbf{v}}{(\imath c-\boldsymbol{\imath}\cdot\mathbf{v})}=\frac{\mathbf{v}}{c^2}V(\mathbf{r},t).$
10.3.2 the fields of a moving point charge	
E of a moving point	We have the Liénard-Wiechert potentials: $$V(\mathbf{r},t)=\frac{1}{4\pi\epsilon_0}\frac{qc}{(\imath c-\boldsymbol{\imath}\cdot\mathbf{v})},\qquad\mathbf{A}(\mathbf{r},t)=\frac{\mathbf{v}}{c^2}V(\mathbf{r},t),$$ and we can find E and B via: $$\mathbf{E}=-\nabla V-\frac{\partial\mathbf{A}}{\partial t},\qquad\mathbf{B}=\nabla\times\mathbf{A}.$$ we have to consider however: $$\boldsymbol{\imath}=\mathbf{r}-\mathbf{w}(t_r)\quad\text{and}\quad\mathbf{v}=\dot{\mathbf{w}}(t_r)$$ with: $|\mathbf{r}-\mathbf{w}(t_r)|=c(t-t_r)$ Then the electric field is: $$\mathbf{E}(\mathbf{r},t)=\frac{q}{4\pi\epsilon_0}\frac{\imath}{(\boldsymbol{\imath}\cdot\mathbf{u})^3}\left[(c^2-v^2)\mathbf{u}+\boldsymbol{\imath}\times(\mathbf{u}\times\mathbf{a})\right].$$!!proof see p463!!
B of a moving point	Using the formulas from above, we find: $$\mathbf{B}(\mathbf{r},t)=\frac{1}{c}\hat{\boldsymbol{\imath}}\times\mathbf{E}(\mathbf{r},t).$$ proof p464